

STEAM-AI Diagnostic Report

SCEN_2030, horizon 2030, run SCEN_2030_2030_57801c41

Generated 2026-09-22T22:25:51+00:00 by STEAM-AI 0.1.0. Ingested 2026-09-22T21:06:09.892212+00:00.

DRAFT: DRAFT - generated by STEAM-AI for modeller review

SYNTHETIC DATA: SYNTHETIC DATA - this run was generated for testing, not a STEAM output

Run details

Item	Value
Run id	SCEN_2030_2030_57801c41
Scenario	SCEN_2030
Horizon year	2030
Policy set	REF
STEAM version	synthetic-0.1
Ingested at	2026-09-22T21:06:09.892212+00:00
Report generated at	2026-09-22T22:25:51+00:00
STEAM-AI version	0.1.0

[src: manifest.json]

This report lists what STEAM-AI found when it checked the run, ranked by how much each issue matters. Every number carries a source reference in small grey text. The report is a draft for modeller review and is not a model result.

Executive summary

Run health score: 71.3 out of 100, grade C.

[src: health.json]

Definition: Health = 0.7 x Findings component + 0.3 x Convergence component. Findings component = $\max(0, 100 - 25 \times \text{Critical} - 8 \times \text{High} - 2 \times \text{Medium} - 0 \times \text{Info})$, counting each distinct problem type (a check and the issue it found) once, at its worst significant severity: a coding error repeated on 8,000 links is one problem. Convergence component = 100 if every assignment period met the relative gap target, scaled down linearly to 0 at ten times the target. When a run has no convergence data, the findings component carries the full weight. Grade: A ≥ 90 , B ≥ 75 , C ≥ 60 , D ≥ 40 , E below 40.

[src: config/health.yaml; health.json]

Health score components

Component	Score	Weight	Detail
Findings	59.0	0.70	100 - 41 penalty over 4 distinct problem type(s) across 41 significant finding(s) (Critical 10, High 11, Medium 19, Info 1)
Convergence	100.0	0.30	target 0.001; per period: HWY_ASSIGN AM: gap 0.0005 -> 100; HWY_ASSIGN EV: gap 0.0005 -> 100; HWY_ASSIGN MD: gap 0.0005 -> 100; HWY_ASSIGN NT: gap 0.0005 -> 100; HWY_ASSIGN PM: gap 0.0005 -> 100

[src: health.json]

The checks produced 41 findings: 10 Critical, 11 High, 19 Medium, 1 Info.

[src: findings.json]

Findings by severity

Severity	Findings	Significant
Critical	10	10
High	11	11
Medium	19	19
Info	1	1

[src: findings.json]

Top five findings

Top five findings by severity

Rank	Severity	Location	Finding
1	Critical	Link 199 (FWY, 1107-1108), AM	Link 199 (FWY) carries 37,440 vehicles in the AM peak, 1.56 times its capacity.
2	Critical	Link 191 (FWY, 1103-1104), AM	Link 191 (FWY) carries 35,520 vehicles in the AM peak, 1.48 times its capacity.
3	Critical	Link 193 (FWY, 1104-1105), AM	Link 193 (FWY) carries 36,000 vehicles in the AM peak, 1.50 times its capacity.
4	Critical	Link 189 (FWY, 1102-1103), AM	Link 189 (FWY) carries 35,040 vehicles in the AM peak, 1.46 times its capacity.
5	Critical	Link 197 (FWY, 1106-1107), AM	Link 197 (FWY) carries 36,960 vehicles in the AM peak, 1.54 times its capacity.

[src: findings/24992b6a31d7; findings/3cfc28052b56; findings/3e5c6f481bed; findings/65aa8257d2d8; findings/81d9e99d5ab0]

What to fix first

21 Critical or High findings need action. Fix them in the order below; Critical items block use of the run.

[src: findings.json]

Action list

Severity	Location	Finding	Suggested action	Best measure
Critical	Link 199 (FWY, 1107-1108), AM	Link 199 (FWY) carries 37,440 vehicles in the AM peak, 1.56 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.56 to about 1.25 if demand is unchanged
Critical	Link 191 (FWY, 1103-1104), AM	Link 191 (FWY) carries 35,520 vehicles in the AM peak, 1.48 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.48 to about 1.18 if demand is unchanged
Critical	Link 193 (FWY, 1104-1105), AM	Link 193 (FWY) carries 36,000 vehicles in the AM peak, 1.50 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.50 to about 1.20 if demand is unchanged
Critical	Link 189 (FWY, 1102-1103), AM	Link 189 (FWY) carries 35,040 vehicles in the AM peak, 1.46 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.46 to about 1.17 if demand is unchanged
Critical	Link 197 (FWY, 1106-1107), AM	Link 197 (FWY) carries 36,960 vehicles in the AM peak, 1.54 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.54 to about 1.23 if demand is unchanged
Critical	Link 185 (FWY, 1100-1101), AM	Link 185 (FWY) carries 34,080 vehicles in the AM peak, 1.42 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.42 to about 1.14 if demand is unchanged
Critical	Link 201 (FWY, 1108-1109), AM	Link 201 (FWY) carries 37,920 vehicles in the AM peak, 1.58 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.58 to about 1.26 if demand is unchanged
Critical	Link 183 (FWY, 1099-1100), AM	Link 183 (FWY) carries 33,600 vehicles in the AM peak, 1.40 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.40 to about 1.12 if demand is unchanged
Critical	Link 187 (FWY, 1101-1102), AM	Link 187 (FWY) carries 34,560 vehicles in the AM peak, 1.44 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.44 to about 1.15 if demand is unchanged
Critical	Link 195 (FWY, 1105-1106), AM	Link 195 (FWY) carries 36,480 vehicles in the AM peak, 1.52 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.52 to about 1.22 if demand is unchanged

[src: findings/24992b6a31d7; findings/3cfc28052b56; findings/3e5c6f481bed; findings/65aa8257d2d8; findings/81d9e99d5ab0; findings/95aff43d2c28; findings/9914fb4f0d20; findings/baafae99d76e; findings/db859fde7a54; findings/de5af663021e]

11 further Critical or High findings are listed in the findings section.

[src: findings.json]

Findings by severity

Critical findings (10)

Link volume outliers: Link 199 (FWY, 1107-1108), AM

Critical: Link 199 (FWY) carries 37,440 vehicles in the AM peak, 1.56 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=3.744e+04, vc_ratio=1.56, zscore=9.365, group_median_volume=9246, group_size=24, link_class=FWY, area_type=CBD, capacity_vph=8000, lanes=4, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3103

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/24992b6a31d7; link_flows.csv (link_flows) row 3103 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/24992b6a31d7; link_flows.csv (link_flows) row 3103 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/24992b6a31d7; link_flows.csv (link_flows) row 3103 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	37,440	
vc_ratio	1.56	
zscore	9.365	
group_median_volume	9,250	
group_size	24	
link_class	FWY	
area_type	CBD	
capacity_vph	8,000	
lanes	4	
too_low	no	
vc_critical		1.30

Metric	Value	Threshold
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/24992b6a31d7; link_flows.csv (link_flows) row 3103 col volume]

Sources: link_flows.csv (link_flows) row 3103 col volume

[src: findings/24992b6a31d7; link_flows.csv (link_flows) row 3103 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 8,000 to 10,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.56 to about 1.25 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/24992b6a31d7; link_flows.csv (link_flows) row 3103 col volume]

Link volume outliers: Link 191 (FWY, 1103-1104), AM

Critical: Link 191 (FWY) carries 35,520 vehicles in the AM peak, 1.48 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=3.552e+04, vc_ratio=1.48, zscore=9.013, group_median_volume=9246, group_size=24, link_class=FWY, area_type=CBD, capacity_vph=8000, lanes=4, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3095

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/3cfc28052b56; link_flows.csv (link_flows) row 3095 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/3cfc28052b56; link_flows.csv (link_flows) row 3095 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/3cfc28052b56; link_flows.csv (link_flows) row 3095 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	35,520	
vc_ratio	1.48	
zscore	9.013	
group_median_volume	9,250	
group_size	24	
link_class	FWY	
area_type	CBD	
capacity_vph	8,000	
lanes	4	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/3cfc28052b56; link_flows.csv (link_flows) row 3095 col volume]

Sources: link_flows.csv (link_flows) row 3095 col volume

[src: findings/3cfc28052b56; link_flows.csv (link_flows) row 3095 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 8,000 to 10,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.48 to about 1.18 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/3cfc28052b56; link_flows.csv (link_flows) row 3095 col volume]

Link volume outliers: Link 193 (FWY, 1104-1105), AM

Critical: Link 193 (FWY) carries 36,000 vehicles in the AM peak, 1.50 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=3.6e+04, vc_ratio=1.5, zscore=9.102, group_median_volume=9246, group_size=24, link_class=FWY, area_type=CBD, capacity_vph=8000, lanes=4, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3097

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/3e5c6f481bed; link_flows.csv (link_flows) row 3097 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/3e5c6f481bed; link_flows.csv (link_flows) row 3097 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/3e5c6f481bed; link_flows.csv (link_flows) row 3097 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	36,000	
vc_ratio	1.50	
zscore	9.102	
group_median_volume	9,250	
group_size	24	
link_class	FWY	
area_type	CBD	
capacity_vph	8,000	
lanes	4	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/3e5c6f481bed; link_flows.csv (link_flows) row 3097 col volume]

Sources: link_flows.csv (link_flows) row 3097 col volume

[src: findings/3e5c6f481bed; link_flows.csv (link_flows) row 3097 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 8,000 to 10,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.50 to about 1.20 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/3e5c6f481bed; link_flows.csv (link_flows) row 3097 col volume]

Link volume outliers: Link 189 (FWY, 1102-1103), AM

Critical: Link 189 (FWY) carries 35,040 vehicles in the AM peak, 1.46 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=3.504e+04, vc_ratio=1.46, zscore=9.279, group_median_volume=6522, group_size=40, link_class=FWY, area_type=URB, capacity_vph=8000, lanes=4, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3093

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/65aa8257d2d8; link_flows.csv (link_flows) row 3093 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/65aa8257d2d8; link_flows.csv (link_flows) row 3093 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/65aa8257d2d8; link_flows.csv (link_flows) row 3093 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	35,040	
vc_ratio	1.46	
zscore	9.279	
group_median_volume	6,520	

Metric	Value	Threshold
group_size	40	
link_class	FWY	
area_type	URB	
capacity_vph	8,000	
lanes	4	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/65aa8257d2d8; link_flows.csv (link_flows) row 3093 col volume]

Sources: link_flows.csv (link_flows) row 3093 col volume

[src: findings/65aa8257d2d8; link_flows.csv (link_flows) row 3093 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 8,000 to 10,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.46 to about 1.17 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/65aa8257d2d8; link_flows.csv (link_flows) row 3093 col volume]

Link volume outliers: Link 197 (FWY, 1106-1107), AM

Critical: Link 197 (FWY) carries 36,960 vehicles in the AM peak, 1.54 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=3.696e+04, vc_ratio=1.54, zscore=9.279, group_median_volume=9246, group_size=24, link_class=FWY, area_type=CBD, capacity_vph=8000, lanes=4, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3101

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS

mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/81d9e99d5ab0; link_flows.csv (link_flows) row 3101 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/81d9e99d5ab0; link_flows.csv (link_flows) row 3101 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/81d9e99d5ab0; link_flows.csv (link_flows) row 3101 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	36,960	
vc_ratio	1.54	
zscore	9.279	
group_median_volume	9,250	
group_size	24	
link_class	FWY	
area_type	CBD	
capacity_vph	8,000	
lanes	4	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/81d9e99d5ab0; link_flows.csv (link_flows) row 3101 col volume]

Sources: link_flows.csv (link_flows) row 3101 col volume

[src: findings/81d9e99d5ab0; link_flows.csv (link_flows) row 3101 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 8,000 to 10,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.54 to about 1.23 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/81d9e99d5ab0; link_flows.csv (link_flows) row 3101 col volume]

Link volume outliers: Link 185 (FWY, 1100-1101), AM

Critical: Link 185 (FWY) carries 34,080 vehicles in the AM peak, 1.42 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=3.408e+04, vc_ratio=1.42, zscore=12.18, group_median_volume=4897, group_size=84, link_class=FWY, area_type=SUB, capacity_vph=8000, lanes=4, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3089

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/95aff43d2c28; link_flows.csv (link_flows) row 3089 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/95aff43d2c28; link_flows.csv (link_flows) row 3089 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/95aff43d2c28; link_flows.csv (link_flows) row 3089 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	34,080	
vc_ratio	1.42	
zscore	12.2	
group_median_volume	4,900	
group_size	84	
link_class	FWY	
area_type	SUB	
capacity_vph	8,000	
lanes	4	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3

Metric	Value	Threshold
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/95aff43d2c28; link_flows.csv (link_flows) row 3089 col volume]

Sources: link_flows.csv (link_flows) row 3089 col volume

[src: findings/95aff43d2c28; link_flows.csv (link_flows) row 3089 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 8,000 to 10,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.42 to about 1.14 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/95aff43d2c28; link_flows.csv (link_flows) row 3089 col volume]

Link volume outliers: Link 201 (FWY, 1108-1109), AM

Critical: Link 201 (FWY) carries 37,920 vehicles in the AM peak, 1.58 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=3.792e+04, vc_ratio=1.58, zscore=9.715, group_median_volume=6522, group_size=40, link_class=FWY, area_type=URB, capacity_vph=8000, lanes=4, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3105

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/9914fb4f0d20; link_flows.csv (link_flows) row 3105 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/9914fb4f0d20; link_flows.csv (link_flows) row 3105 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/9914fb4f0d20; link_flows.csv (link_flows) row 3105 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	37,920	
vc_ratio	1.58	
zscore	9.715	
group_median_volume	6,520	
group_size	40	
link_class	FWY	
area_type	URB	
capacity_vph	8,000	
lanes	4	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/9914fb4f0d20; link_flows.csv (link_flows) row 3105 col volume]

Sources: link_flows.csv (link_flows) row 3105 col volume

[src: findings/9914fb4f0d20; link_flows.csv (link_flows) row 3105 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 8,000 to 10,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.58 to about 1.26 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/9914fb4f0d20; link_flows.csv (link_flows) row 3105 col volume]

Link volume outliers: Link 183 (FWY, 1099-1100), AM

Critical: Link 183 (FWY) carries 33,600 vehicles in the AM peak, 1.40 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=3.36e+04, vc_ratio=1.4, zscore=12.09, group_median_volume=4897, group_size=84, link_class=FWY, area_type=SUB, capacity_vph=8000, lanes=4, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3087

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period,

```
f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class,
l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING
(link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND
l.link_class NOT IN ('CONN') ), g AS ( SELECT link_class, area_type, period, COUNT(*) AS n,
MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3 ), d AS
( SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING
(link_class, area_type, period) ), m AS ( SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS
mad FROM d GROUP BY 1, 2, 3 ), scored AS ( SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad
> 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol >
0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period) )
SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0)
>= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST
```

[src: findings/baafae99d76e; link_flows.csv (link_flows) row 3087 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/baafae99d76e; link_flows.csv (link_flows) row 3087 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/baafae99d76e; link_flows.csv (link_flows) row 3087 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	33,600	
vc_ratio	1.40	
zscore	12.1	
group_median_volume	4,900	
group_size	84	
link_class	FWY	
area_type	SUB	
capacity_vph	8,000	
lanes	4	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/baafae99d76e; link_flows.csv (link_flows) row 3087 col volume]

Sources: link_flows.csv (link_flows) row 3087 col volume

[src: findings/baafae99d76e; link_flows.csv (link_flows) row 3087 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 8,000 to 10,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.40 to about 1.12 if demand is unchanged	sketch elasticity	medium

Measure	Description	Estimated effect	Method	Confidence
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/baafae99d76e; link_flows.csv (link_flows) row 3087 col volume]

Link volume outliers: Link 187 (FWY, 1101-1102), AM

Critical: Link 187 (FWY) carries 34,560 vehicles in the AM peak, 1.44 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=3.456e+04, vc_ratio=1.44, zscore=9.203, group_median_volume=6522, group_size=40, link_class=FWY, area_type=URB, capacity_vph=8000, lanes=4, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3091

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/db859fde7a54; link_flows.csv (link_flows) row 3091 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/db859fde7a54; link_flows.csv (link_flows) row 3091 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/db859fde7a54; link_flows.csv (link_flows) row 3091 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	34,560	
vc_ratio	1.44	
zscore	9.203	
group_median_volume	6,520	
group_size	40	
link_class	FWY	
area_type	URB	

Metric	Value	Threshold
capacity_vph	8,000	
lanes	4	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/db859fde7a54; link_flows.csv (link_flows) row 3091 col volume]

Sources: link_flows.csv (link_flows) row 3091 col volume

[src: findings/db859fde7a54; link_flows.csv (link_flows) row 3091 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 8,000 to 10,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.44 to about 1.15 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/db859fde7a54; link_flows.csv (link_flows) row 3091 col volume]

Link volume outliers: Link 195 (FWY, 1105-1106), AM

Critical: Link 195 (FWY) carries 36,480 vehicles in the AM peak, 1.52 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=3.648e+04, vc_ratio=1.52, zscore=9.191, group_median_volume=9246, group_size=24, link_class=FWY, area_type=CBD, capacity_vph=8000, lanes=4, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3099

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period))

SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/de5af663021e; link_flows.csv (link_flows) row 3099 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/de5af663021e; link_flows.csv (link_flows) row 3099 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/de5af663021e; link_flows.csv (link_flows) row 3099 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	36,480	
vc_ratio	1.52	
zscore	9.191	
group_median_volume	9,250	
group_size	24	
link_class	FWY	
area_type	CBD	
capacity_vph	8,000	
lanes	4	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/de5af663021e; link_flows.csv (link_flows) row 3099 col volume]

Sources: link_flows.csv (link_flows) row 3099 col volume

[src: findings/de5af663021e; link_flows.csv (link_flows) row 3099 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 8,000 to 10,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.52 to about 1.22 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/de5af663021e; link_flows.csv (link_flows) row 3099 col volume]

High findings (11)

Link volume outliers: Link 394 (LOC, 1030-1016), AM

High: Link 394 (LOC, RUR) carries 993 vehicles in the AM peak, far above similar links (typical 56).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=993.2, vc_ratio=0.6621, zscore=4.423, group_median_volume=56.2, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3298

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/0be2badc3729; link_flows.csv (link_flows) row 3298 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/0be2badc3729; link_flows.csv (link_flows) row 3298 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/0be2badc3729; link_flows.csv (link_flows) row 3298 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	990	
vc_ratio	0.66	
zscore	4.423	
group_median_volume	60	
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%

Metric	Value	Threshold
min_vc_for_outlier		0.25

[src: findings/0be2badc3729; link_flows.csv (link_flows) row 3298 col volume]

Sources: link_flows.csv (link_flows) row 3298 col volume

[src: findings/0be2badc3729; link_flows.csv (link_flows) row 3298 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Review link attributes against neighbours	Compare class, speed and capacity with adjacent links of the same class; correct any outlier attribute and re-assign.	Not estimated	not computable	low
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/0be2badc3729; link_flows.csv (link_flows) row 3298 col volume]

Trip length distribution: Trip lengths HBW CAR

High: HBW CAR trips average 10,504.5 km, 40.3% longer than the 7,485.2 km in the base run BASE_2025_2025_52d15c27.

Modeller view: Method: trip-weighted mean of DIST skim over DEMAND cells per purpose x mode

Values: mean_km=1.05e+04, mean_km_base=7485, mean_diff_share=0.4034, trips=1.201e+05,

trips_base=1.157e+05, comparison=base run BASE_2025_2025_52d15c27

Thresholds: max_mean_diff_share=0.15, max_mean_diff_share_high=0.3

Files: od.csv; skims.csv

SQL: SELECT o.purpose, o.mode, COUNT(*) AS n_cells, SUM(o.trips) AS trips, SUM(o.trips * s.value * 1.0) / NULLIF(SUM(o.trips), 0) AS mean_km FROM od o JOIN skims s ON s.skim_kind = 'DIST' AND s.mode = o.mode AND s.period = o.period AND s.origin = o.origin AND s.destination = o.destination WHERE o.matrix_kind = 'DEMAND' AND o.trips > 0 GROUP BY 1, 2 ORDER BY 1, 2

[src: findings/8a7950d1465b; od.csv (od) col trips; skims.csv (skims) col value]

Likely cause: Changed distribution parameters, skims (network speeds or costs) or zone system rather than a genuine land-use effect.

[src: findings/8a7950d1465b; od.csv (od) col trips; skims.csv (skims) col value]

Suggested action: Compare the DIST/TIME skims and the distribution parameters with the base before accepting the result.

[src: findings/8a7950d1465b; od.csv (od) col trips; skims.csv (skims) col value]

Evidence

Metric	Value	Threshold
mean_km	10,504	
mean_km_base	7,485	
mean_diff_share	40.3%	
trips	120,130	
trips_base	115,700	
bins_km	8 items (full list in findings.json): 0, 2, 5	
run_share	8 items (full list in findings.json): 0, 0, 0	
base_share	8 items (full list in findings.json): 0, 0, 0	
comparison	base run BASE_2025_2025_52d15c27	
max_mean_diff_share		15.0%
max_mean_diff_share_high		30.0%

[src: findings/8a7950d1465b; od.csv (od) col trips; skims.csv (skims) col value]

Sources: od.csv (od) col trips; skims.csv (skims) col value

[src: findings/8a7950d1465b; od.csv (od) col trips; skims.csv (skims) col value]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Compare skims with the base	Check DIST/TIME skims and distribution parameters against the base run.	Not estimated	not computable	low

[src: findings/8a7950d1465b; od.csv (od) col trips; skims.csv (skims) col value]

Link volume outliers: Link 588 (COL, 1121-1107), PM

High: Link 588 (COL) carries 2,565 vehicles in the PM peak, 1.22 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2565, vc_ratio=1.221, zscore=1.253,

group_median_volume=850.1, group_size=22, link_class=COL, area_type=CBD, capacity_vph=700, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 11236

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/8d04a74eff1c; link_flows.csv (link_flows) row 11236 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/8d04a74eff1c; link_flows.csv (link_flows) row 11236 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/8d04a74eff1c; link_flows.csv (link_flows) row 11236 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	2,570	
vc_ratio	1.22	
zscore	1.253	
group_median_volume	850	
group_size	22	
link_class	COL	
area_type	CBD	

Metric	Value	Threshold
capacity_vph	700	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/8d04a74eff1c; link_flows.csv (link_flows) row 11236 col volume]

Sources: link_flows.csv (link_flows) row 11236 col volume

[src: findings/8d04a74eff1c; link_flows.csv (link_flows) row 11236 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 700 to 1,400 veh/h (one lane = 700 veh/h).	V/C falls from 1.22 to about 0.61 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/8d04a74eff1c; link_flows.csv (link_flows) row 11236 col volume]

Link volume outliers: Link 536 (ART, 1119-1105), PM

High: Link 536 (ART) carries 6,241 vehicles in the PM peak, 1.16 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=6241, vc_ratio=1.156, zscore=1.219, group_median_volume=1811, group_size=30, link_class=ART, area_type=CBD, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 11184

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0))

≥ 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/957dfa057e28; link_flows.csv (link_flows) row 11184 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/957dfa057e28; link_flows.csv (link_flows) row 11184 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/957dfa057e28; link_flows.csv (link_flows) row 11184 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	6,240	
vc_ratio	1.16	
zscore	1.219	
group_median_volume	1,810	
group_size	30	
link_class	ART	
area_type	CBD	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/957dfa057e28; link_flows.csv (link_flows) row 11184 col volume]

Sources: link_flows.csv (link_flows) row 11184 col volume

[src: findings/957dfa057e28; link_flows.csv (link_flows) row 11184 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 1,800 to 2,700 veh/h (one lane = 900 veh/h).	V/C falls from 1.16 to about 0.77 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/957dfa057e28; link_flows.csv (link_flows) row 11184 col volume]

Link volume outliers: Link 726 (LOC, 1182-1168), AM

High: Link 726 (LOC, RUR) carries 798 vehicles in the AM peak, far above similar links (typical 56).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=797.9, vc_ratio=0.5319, zscore=4.084, group_median_volume=56.2, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3630

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/b0d7c4c9c4d8; link_flows.csv (link_flows) row 3630 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/b0d7c4c9c4d8; link_flows.csv (link_flows) row 3630 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/b0d7c4c9c4d8; link_flows.csv (link_flows) row 3630 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	800	
vc_ratio	0.53	
zscore	4.084	
group_median_volume	60	
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/b0d7c4c9c4d8; link_flows.csv (link_flows) row 3630 col volume]

Sources: link_flows.csv (link_flows) row 3630 col volume

[src: findings/b0d7c4c9c4d8; link_flows.csv (link_flows) row 3630 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Review link attributes against neighbours	Compare class, speed and capacity with adjacent links of the same class; correct any outlier attribute and re-assign.	Not estimated	not computable	low
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/b0d7c4c9c4d8; link_flows.csv (link_flows) row 3630 col volume]

Link volume outliers: Link 691 (ART, 1111-1125), AM

High: Link 691 (ART, SUB) carries 2,305 vehicles in the AM peak, far above similar links (typical 589).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=2305, vc_ratio=0.4268, zscore=4.503,

group_median_volume=589, group_size=92, link_class=ART, area_type=SUB, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3595

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/bf58897c6416; link_flows.csv (link_flows) row 3595 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/bf58897c6416; link_flows.csv (link_flows) row 3595 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/bf58897c6416; link_flows.csv (link_flows) row 3595 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	2,300	
vc_ratio	0.43	
zscore	4.503	
group_median_volume	590	
group_size	92	
link_class	ART	

Metric	Value	Threshold
area_type	SUB	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/bf58897c6416; link_flows.csv (link_flows) row 3595 col volume]

Sources: link_flows.csv (link_flows) row 3595 col volume

[src: findings/bf58897c6416; link_flows.csv (link_flows) row 3595 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Review link attributes against neighbours	Compare class, speed and capacity with adjacent links of the same class; correct any outlier attribute and re-assign.	Not estimated	not computable	low
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/bf58897c6416; link_flows.csv (link_flows) row 3595 col volume]

Link volume outliers: Link 692 (ART, 1125-1111), AM

High: Link 692 (ART, SUB) carries 2,292 vehicles in the AM peak, far above similar links (typical 589).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=2292, vc_ratio=0.4244, zscore=4.485, group_median_volume=589, group_size=92, link_class=ART, area_type=SUB, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3596

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/c638a1090952; link_flows.csv (link_flows) row 3596 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/c638a1090952; link_flows.csv (link_flows) row 3596 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/c638a1090952; link_flows.csv (link_flows) row 3596 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	2,290	
vc_ratio	0.42	
zscore	4.485	
group_median_volume	590	
group_size	92	
link_class	ART	
area_type	SUB	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/c638a1090952; link_flows.csv (link_flows) row 3596 col volume]

Sources: link_flows.csv (link_flows) row 3596 col volume

[src: findings/c638a1090952; link_flows.csv (link_flows) row 3596 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Review link attributes against neighbours	Compare class, speed and capacity with adjacent links of the same class; correct any outlier attribute and re-assign.	Not estimated	not computable	low
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/c638a1090952; link_flows.csv (link_flows) row 3596 col volume]

Link volume outliers: Link 706 (LOC, 1042-1028), AM

High: Link 706 (LOC, RUR) carries 791 vehicles in the AM peak, far above similar links (typical 56).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=791, vc_ratio=0.5273, zscore=4.07, group_median_volume=56.2, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3610

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/cb2811b72972; link_flows.csv (link_flows) row 3610 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/cb2811b72972; link_flows.csv (link_flows) row 3610 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/cb2811b72972; link_flows.csv (link_flows) row 3610 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	790	
vc_ratio	0.53	
zscore	4.07	
group_median_volume	60	
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/cb2811b72972; link_flows.csv (link_flows) row 3610 col volume]

Sources: link_flows.csv (link_flows) row 3610 col volume

[src: findings/cb2811b72972; link_flows.csv (link_flows) row 3610 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Review link attributes against neighbours	Compare class, speed and capacity with adjacent links of the same class; correct any outlier attribute and re-assign.	Not estimated	not computable	low

Measure	Description	Estimated effect	Method	Confidence
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/cb2811b72972; link_flows.csv (link_flows) row 3610 col volume]

Link volume outliers: Link 705 (LOC, 1028-1042), AM

High: Link 705 (LOC, RUR) carries 913 vehicles in the AM peak, far above similar links (typical 56).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=913.2, vc_ratio=0.6088, zscore=4.293, group_median_volume=56.2, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3609

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/ccd4663a9044; link_flows.csv (link_flows) row 3609 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/ccd4663a9044; link_flows.csv (link_flows) row 3609 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/ccd4663a9044; link_flows.csv (link_flows) row 3609 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	910	
vc_ratio	0.61	
zscore	4.293	
group_median_volume	60	
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	

Metric	Value	Threshold
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/ccd4663a9044; link_flows.csv (link_flows) row 3609 col volume]

Sources: link_flows.csv (link_flows) row 3609 col volume

[src: findings/ccd4663a9044; link_flows.csv (link_flows) row 3609 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Review link attributes against neighbours	Compare class, speed and capacity with adjacent links of the same class; correct any outlier attribute and re-assign.	Not estimated	not computable	low
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/ccd4663a9044; link_flows.csv (link_flows) row 3609 col volume]

Link volume outliers: Link 393 (LOC, 1016-1030), AM

High: Link 393 (LOC, RUR) carries 1,097 vehicles in the AM peak, far above similar links (typical 56).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=1097, vc_ratio=0.7311, zscore=4.576, group_median_volume=56.2, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3297

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/ec37afcb6720; link_flows.csv (link_flows) row 3297 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/ec37afcb6720; link_flows.csv (link_flows) row 3297 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/ec37afcb6720; link_flows.csv (link_flows) row 3297 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	1,100	
vc_ratio	0.73	
zscore	4.576	
group_median_volume	60	
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/ec37afcb6720; link_flows.csv (link_flows) row 3297 col volume]

Sources: link_flows.csv (link_flows) row 3297 col volume

[src: findings/ec37afcb6720; link_flows.csv (link_flows) row 3297 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Review link attributes against neighbours	Compare class, speed and capacity with adjacent links of the same class; correct any outlier attribute and re-assign.	Not estimated	not computable	low
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/ec37afcb6720; link_flows.csv (link_flows) row 3297 col volume]

Link volume outliers: Link 587 (COL, 1107-1121), AM

High: Link 587 (COL) carries 2,380 vehicles in the AM peak, 1.13 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2380, vc_ratio=1.133, zscore=1.471,

group_median_volume=632.4, group_size=22, link_class=COL, area_type=CBD, capacity_vph=700, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3491

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n,

```

MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3 ), d AS
( SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING
(link_class, area_type, period) ), m AS ( SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS
mad FROM d GROUP BY 1, 2, 3 ), scored AS ( SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad
> 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol >
0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period) )
SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0)
>= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

```

[src: findings/f5416817f042; link_flows.csv (link_flows) row 3491 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/f5416817f042; link_flows.csv (link_flows) row 3491 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/f5416817f042; link_flows.csv (link_flows) row 3491 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	2,380	
vc_ratio	1.13	
zscore	1.471	
group_median_volume	630	
group_size	22	
link_class	COL	
area_type	CBD	
capacity_vph	700	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/f5416817f042; link_flows.csv (link_flows) row 3491 col volume]

Sources: link_flows.csv (link_flows) row 3491 col volume

[src: findings/f5416817f042; link_flows.csv (link_flows) row 3491 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 700 to 1,400 veh/h (one lane = 700 veh/h).	V/C falls from 1.13 to about 0.57 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium

Measure	Description	Estimated effect	Method	Confidence
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/f5416817f042; link_flows.csv (link_flows) row 3491 col volume]

Medium findings (19)

Link volume outliers: Link 315 (ART, 1170-1171), AM

Medium: Link 315 (ART, RUR) carries 2,340 vehicles in the AM peak, far above similar links (typical 611).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=2340, vc_ratio=0.4333, zscore=3.75, group_median_volume=611.3, group_size=58, link_class=ART, area_type=RUR, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3219

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/2b385e1a2195; link_flows.csv (link_flows) row 3219 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/2b385e1a2195; link_flows.csv (link_flows) row 3219 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/2b385e1a2195; link_flows.csv (link_flows) row 3219 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	2,340	
vc_ratio	0.43	
zscore	3.75	
group_median_volume	610	
group_size	58	
link_class	ART	
area_type	RUR	
capacity_vph	1,800	

Metric	Value	Threshold
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/2b385e1a2195; link_flows.csv (link_flows) row 3219 col volume]

Sources: link_flows.csv (link_flows) row 3219 col volume

[src: findings/2b385e1a2195; link_flows.csv (link_flows) row 3219 col volume]

Link volume outliers: Link 689 (ART, 1097-1111), AM

Medium: Link 689 (ART, SUB) carries 1,489 vehicles in the AM peak, far above similar links (typical 589).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=1489, vc_ratio=0.2758, zscore=3.062, group_median_volume=589, group_size=92, link_class=ART, area_type=SUB, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3593

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/33405672a01d; link_flows.csv (link_flows) row 3593 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/33405672a01d; link_flows.csv (link_flows) row 3593 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/33405672a01d; link_flows.csv (link_flows) row 3593 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	1,490	
vc_ratio	0.28	

Metric	Value	Threshold
zscore	3.062	
group_median_volume	590	
group_size	92	
link_class	ART	
area_type	SUB	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/33405672a01d; link_flows.csv (link_flows) row 3593 col volume]

Sources: link_flows.csv (link_flows) row 3593 col volume

[src: findings/33405672a01d; link_flows.csv (link_flows) row 3593 col volume]

Link volume outliers: Link 715 (LOC, 1098-1112), AM

Medium: Link 715 (LOC, RUR) carries 578 vehicles in the AM peak, far above similar links (typical 56).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=577.9, vc_ratio=0.3853, zscore=3.585, group_median_volume=56.2, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3619

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/49de25648af3; link_flows.csv (link_flows) row 3619 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/49de25648af3; link_flows.csv (link_flows) row 3619 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/49de25648af3; link_flows.csv (link_flows) row 3619 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	580	
vc_ratio	0.39	
zscore	3.585	
group_median_volume	60	
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/49de25648af3; link_flows.csv (link_flows) row 3619 col volume]

Sources: link_flows.csv (link_flows) row 3619 col volume

[src: findings/49de25648af3; link_flows.csv (link_flows) row 3619 col volume]

Link volume outliers: Link 446 (ART, 1032-1018), AM**Medium:** Link 446 (ART, SUB) carries 1,528 vehicles in the AM peak, far above similar links (typical 589).Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=1528, vc_ratio=0.283, zscore=3.146, group_median_volume=589, group_size=92, link_class=ART, area_type=SUB, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3350

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/4f0f5f8fcf24; link_flows.csv (link_flows) row 3350 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/4f0f5f8fcf24; link_flows.csv (link_flows) row 3350 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/4f0f5f8fcf24; link_flows.csv (link_flows) row 3350 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	1,530	
vc_ratio	0.28	
zscore	3.146	
group_median_volume	590	
group_size	92	
link_class	ART	
area_type	SUB	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/4f0f5f8fcf24; link_flows.csv (link_flows) row 3350 col volume]

Sources: link_flows.csv (link_flows) row 3350 col volume

[src: findings/4f0f5f8fcf24; link_flows.csv (link_flows) row 3350 col volume]

Link volume outliers: Link 687 (ART, 1083-1097), AM

Medium: Link 687 (ART, SUB) carries 1,623 vehicles in the AM peak, far above similar links (typical 589).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=1623, vc_ratio=0.3005, zscore=3.345,

group_median_volume=589, group_size=92, link_class=ART, area_type=SUB, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3591

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING

(link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/5763a3a48205; link_flows.csv (link_flows) row 3591 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/5763a3a48205; link_flows.csv (link_flows) row 3591 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/5763a3a48205; link_flows.csv (link_flows) row 3591 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	1,620	
vc_ratio	0.30	
zscore	3.345	
group_median_volume	590	
group_size	92	
link_class	ART	
area_type	SUB	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/5763a3a48205; link_flows.csv (link_flows) row 3591 col volume]

Sources: link_flows.csv (link_flows) row 3591 col volume

[src: findings/5763a3a48205; link_flows.csv (link_flows) row 3591 col volume]

Link volume outliers: Link 466 (ART, 1172-1158), AM

Medium: Link 466 (ART, SUB) carries 1,705 vehicles in the AM peak, far above similar links (typical 589).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=1705, vc_ratio=0.3157, zscore=3.507, group_median_volume=589, group_size=92, link_class=ART, area_type=SUB, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3370

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period,

```
f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class,
l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING
(link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND
l.link_class NOT IN ('CONN') ), g AS ( SELECT link_class, area_type, period, COUNT(*) AS n,
MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3 ), d AS
( SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING
(link_class, area_type, period) ), m AS ( SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS
mad FROM d GROUP BY 1, 2, 3 ), scored AS ( SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad
> 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol >
0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period) )
SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0)
>= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST
```

[src: findings/7fda3454981d; link_flows.csv (link_flows) row 3370 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/7fda3454981d; link_flows.csv (link_flows) row 3370 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/7fda3454981d; link_flows.csv (link_flows) row 3370 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	1,700	
vc_ratio	0.32	
zscore	3.507	
group_median_volume	590	
group_size	92	
link_class	ART	
area_type	SUB	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/7fda3454981d; link_flows.csv (link_flows) row 3370 col volume]

Sources: link_flows.csv (link_flows) row 3370 col volume

[src: findings/7fda3454981d; link_flows.csv (link_flows) row 3370 col volume]

Link volume outliers: Link 692 (ART, 1125-1111), PM

Medium: Link 692 (ART, SUB) carries 2,230 vehicles in the PM peak, far above similar links (typical 633).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=2230, vc_ratio=0.413, zscore=3.068,

group_median_volume=633.2, group_size=92, link_class=ART, area_type=SUB, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 11340

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/88132cae4f3d; link_flows.csv (link_flows) row 11340 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/88132cae4f3d; link_flows.csv (link_flows) row 11340 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/88132cae4f3d; link_flows.csv (link_flows) row 11340 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	2,230	
vc_ratio	0.41	
zscore	3.068	
group_median_volume	630	
group_size	92	
link_class	ART	
area_type	SUB	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/88132cae4f3d; link_flows.csv (link_flows) row 11340 col volume]

Sources: link_flows.csv (link_flows) row 11340 col volume

[src: findings/88132cae4f3d; link_flows.csv (link_flows) row 11340 col volume]

Link volume outliers: Link 601 (ART, 1024-1038), AM

Medium: Link 601 (ART, SUB) carries 1,663 vehicles in the AM peak, far above similar links (typical 589).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=1663, vc_ratio=0.308, zscore=3.426, group_median_volume=589, group_size=92, link_class=ART, area_type=SUB, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3505

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/885561d4043a; link_flows.csv (link_flows) row 3505 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/885561d4043a; link_flows.csv (link_flows) row 3505 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/885561d4043a; link_flows.csv (link_flows) row 3505 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	1,660	
vc_ratio	0.31	
zscore	3.426	
group_median_volume	590	
group_size	92	
link_class	ART	
area_type	SUB	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%

Metric	Value	Threshold
min_vc_for_outlier		0.25

[src: findings/885561d4043a; link_flows.csv (link_flows) row 3505 col volume]

Sources: link_flows.csv (link_flows) row 3505 col volume

[src: findings/885561d4043a; link_flows.csv (link_flows) row 3505 col volume]

Link volume outliers: Link 535 (ART, 1105-1119), AM

Medium: Link 535 (ART) carries 5,417 vehicles in the AM peak, 1.00 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=5417, vc_ratio=1.003, zscore=1.031, group_median_volume=1724, group_size=30, link_class=ART, area_type=CBD, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3439

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/8a1351e9d62a; link_flows.csv (link_flows) row 3439 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/8a1351e9d62a; link_flows.csv (link_flows) row 3439 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/8a1351e9d62a; link_flows.csv (link_flows) row 3439 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	5,420	
vc_ratio	1.00	
zscore	1.031	
group_median_volume	1,720	
group_size	30	
link_class	ART	
area_type	CBD	
capacity_vph	1,800	
lanes	2	
too_low	no	

Metric	Value	Threshold
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/8a1351e9d62a; link_flows.csv (link_flows) row 3439 col volume]

Sources: link_flows.csv (link_flows) row 3439 col volume

[src: findings/8a1351e9d62a; link_flows.csv (link_flows) row 3439 col volume]

Link volume outliers: Link 316 (ART, 1171-1170), AM

Medium: Link 316 (ART, RUR) carries 1,927 vehicles in the AM peak, far above similar links (typical 611).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=1927, vc_ratio=0.3569, zscore=3.209, group_median_volume=611.3, group_size=58, link_class=ART, area_type=RUR, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3220

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/995fc2d532bb; link_flows.csv (link_flows) row 3220 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/995fc2d532bb; link_flows.csv (link_flows) row 3220 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/995fc2d532bb; link_flows.csv (link_flows) row 3220 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	1,930	
vc_ratio	0.36	
zscore	3.209	

Metric	Value	Threshold
group_median_volume	610	
group_size	58	
link_class	ART	
area_type	RUR	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/995fc2d532bb; link_flows.csv (link_flows) row 3220 col volume]

Sources: link_flows.csv (link_flows) row 3220 col volume

[src: findings/995fc2d532bb; link_flows.csv (link_flows) row 3220 col volume]

Link volume outliers: Link 587 (COL, 1107-1121), PM

Medium: Link 587 (COL) carries 2,157 vehicles in the PM peak, 1.03 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2157, vc_ratio=1.027, zscore=1.059,

group_median_volume=850.1, group_size=22, link_class=COL, area_type=CBD, capacity_vph=700, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 11235

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/a68d84d9aeef; link_flows.csv (link_flows) row 11235 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/a68d84d9aeef; link_flows.csv (link_flows) row 11235 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/a68d84d9aeef; link_flows.csv (link_flows) row 11235 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	2,160	
vc_ratio	1.03	
zscore	1.059	
group_median_volume	850	
group_size	22	
link_class	COL	
area_type	CBD	
capacity_vph	700	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/a68d84d9aefe; link_flows.csv (link_flows) row 11235 col volume]

Sources: link_flows.csv (link_flows) row 11235 col volume

[src: findings/a68d84d9aefe; link_flows.csv (link_flows) row 11235 col volume]

Link volume outliers: Link 724 (LOC, 1168-1154), AM

Medium: Link 724 (LOC, RUR) carries 434 vehicles in the AM peak, far above similar links (typical 56).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=434.2, vc_ratio=0.2895, zscore=3.143, group_median_volume=56.2, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3628

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/a8000c88030a; link_flows.csv (link_flows) row 3628 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/a8000c88030a; link_flows.csv (link_flows) row 3628 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/a8000c88030a; link_flows.csv (link_flows) row 3628 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	430	
vc_ratio	0.29	
zscore	3.143	
group_median_volume	60	
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/a8000c88030a; link_flows.csv (link_flows) row 3628 col volume]

Sources: link_flows.csv (link_flows) row 3628 col volume

[src: findings/a8000c88030a; link_flows.csv (link_flows) row 3628 col volume]

Link volume outliers: Link 691 (ART, 1111-1125), PM

Medium: Link 691 (ART, SUB) carries 2,651 vehicles in the PM peak, far above similar links (typical 633).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=2651, vc_ratio=0.4909, zscore=3.49, group_median_volume=633.2, group_size=92, link_class=ART, area_type=SUB, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 11339

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS

mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/be83017bd847; link_flows.csv (link_flows) row 11339 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/be83017bd847; link_flows.csv (link_flows) row 11339 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/be83017bd847; link_flows.csv (link_flows) row 11339 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	2,650	
vc_ratio	0.49	
zscore	3.49	
group_median_volume	630	
group_size	92	
link_class	ART	
area_type	SUB	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/be83017bd847; link_flows.csv (link_flows) row 11339 col volume]

Sources: link_flows.csv (link_flows) row 11339 col volume

[src: findings/be83017bd847; link_flows.csv (link_flows) row 11339 col volume]

Link volume outliers: Link 394 (LOC, 1030-1016), PM

Medium: Link 394 (LOC, RUR) carries 1,218 vehicles in the PM peak, far above similar links (typical 69).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=1218, vc_ratio=0.8123, zscore=3.027, group_median_volume=68.95, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 11042

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class,

```

l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING
(link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND
l.link_class NOT IN ('CONN') ), g AS ( SELECT link_class, area_type, period, COUNT(*) AS n,
MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3 ), d AS
( SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING
(link_class, area_type, period) ), m AS ( SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS
mad FROM d GROUP BY 1, 2, 3 ), scored AS ( SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad
> 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol >
0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period) )
SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0)
>= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

```

[src: findings/c72b14c8cbde; link_flows.csv (link_flows) row 11042 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/c72b14c8cbde; link_flows.csv (link_flows) row 11042 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/c72b14c8cbde; link_flows.csv (link_flows) row 11042 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	1,220	
vc_ratio	0.81	
zscore	3.027	
group_median_volume	70	
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/c72b14c8cbde; link_flows.csv (link_flows) row 11042 col volume]

Sources: link_flows.csv (link_flows) row 11042 col volume

[src: findings/c72b14c8cbde; link_flows.csv (link_flows) row 11042 col volume]

Link volume outliers: Link 588 (COL, 1121-1107), AM

Medium: Link 588 (COL) carries 2,081 vehicles in the AM peak, 0.99 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2081, vc_ratio=0.9909, zscore=1.324,

group_median_volume=632.4, group_size=22, link_class=COL, area_type=CBD, capacity_vph=700,

lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3492

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/df7a6806462a; link_flows.csv (link_flows) row 3492 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/df7a6806462a; link_flows.csv (link_flows) row 3492 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/df7a6806462a; link_flows.csv (link_flows) row 3492 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	2,080	
vc_ratio	0.99	
zscore	1.324	
group_median_volume	630	
group_size	22	
link_class	COL	
area_type	CBD	
capacity_vph	700	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/df7a6806462a; link_flows.csv (link_flows) row 3492 col volume]

Sources: link_flows.csv (link_flows) row 3492 col volume

[src: findings/df7a6806462a; link_flows.csv (link_flows) row 3492 col volume]

Link volume outliers: Link 258 (ART, 1139-1138), AM

Medium: Link 258 (ART, SUB) carries 1,597 vehicles in the AM peak, far above similar links (typical 589).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=1597, vc_ratio=0.2957, zscore=3.291, group_median_volume=589, group_size=92, link_class=ART, area_type=SUB, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3162

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/df9cd89f618a; link_flows.csv (link_flows) row 3162 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/df9cd89f618a; link_flows.csv (link_flows) row 3162 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/df9cd89f618a; link_flows.csv (link_flows) row 3162 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	1,600	
vc_ratio	0.30	
zscore	3.291	
group_median_volume	590	
group_size	92	
link_class	ART	
area_type	SUB	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%

Metric	Value	Threshold
min_vc_for_outlier		0.25

[src: findings/df9cd89f618a; link_flows.csv (link_flows) row 3162 col volume]

Sources: link_flows.csv (link_flows) row 3162 col volume

[src: findings/df9cd89f618a; link_flows.csv (link_flows) row 3162 col volume]

Link volume outliers: Link 725 (LOC, 1168-1182), AM

Medium: Link 725 (LOC, RUR) carries 653 vehicles in the AM peak, far above similar links (typical 56).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=652.9, vc_ratio=0.4353, zscore=3.774, group_median_volume=56.2, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3629

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/e9101c86e625; link_flows.csv (link_flows) row 3629 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/e9101c86e625; link_flows.csv (link_flows) row 3629 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/e9101c86e625; link_flows.csv (link_flows) row 3629 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	650	
vc_ratio	0.44	
zscore	3.774	
group_median_volume	60	
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	

Metric	Value	Threshold
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/e9101c86e625; link_flows.csv (link_flows) row 3629 col volume]

Sources: link_flows.csv (link_flows) row 3629 col volume

[src: findings/e9101c86e625; link_flows.csv (link_flows) row 3629 col volume]

Link volume outliers: Link 716 (LOC, 1112-1098), AM

Medium: Link 716 (LOC, RUR) carries 432 vehicles in the AM peak, far above similar links (typical 56).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=431.7, vc_ratio=0.2878, zscore=3.134, group_median_volume=56.2, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3620

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/eff9181162f4; link_flows.csv (link_flows) row 3620 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/eff9181162f4; link_flows.csv (link_flows) row 3620 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/eff9181162f4; link_flows.csv (link_flows) row 3620 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	430	
vc_ratio	0.29	
zscore	3.134	
group_median_volume	60	

Metric	Value	Threshold
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/eff9181162f4; link_flows.csv (link_flows) row 3620 col volume]

Sources: link_flows.csv (link_flows) row 3620 col volume

[src: findings/eff9181162f4; link_flows.csv (link_flows) row 3620 col volume]

Link volume outliers: Link 706 (LOC, 1042-1028), PM

Medium: Link 706 (LOC, RUR) carries 1,413 vehicles in the PM peak, far above similar links (typical 69).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=1413, vc_ratio=0.9418, zscore=3.184, group_median_volume=68.95, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 11354

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/f8b017d408e3; link_flows.csv (link_flows) row 11354 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/f8b017d408e3; link_flows.csv (link_flows) row 11354 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/f8b017d408e3; link_flows.csv (link_flows) row 11354 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	1,410	
vc_ratio	0.94	
zscore	3.184	
group_median_volume	70	
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/f8b017d408e3; link_flows.csv (link_flows) row 11354 col volume]

Sources: link_flows.csv (link_flows) row 11354 col volume

[src: findings/f8b017d408e3; link_flows.csv (link_flows) row 11354 col volume]

Info findings (1)

Unexplained demand shift: Sector-to-sector demand

Info: 24 sector-to-sector demand shifts above 10% are consistent with input changes in sector S1, S2, S3, S4, S5, S6; the largest is S3 to S3 (34,708 to 27,714 trips).

Modeller view: Method: sector-to-sector DEMAND trips run vs base; shifts explained by changes.sector_change_summary are summarised here

Values: n_explained_shifts=24, metric=trips

Thresholds: sector_shift_share=0.1, min_sector_pair_trips=100, explain_land_use_share=0.01

Files: od.csv

SQL: WITH r AS (SELECT zo.sector_id AS o_sec, zd.sector_id AS d_sec, SUM(o.trips) AS trips FROM od o JOIN zones zo ON zo.zone_id = o.origin JOIN zones zd ON zd.zone_id = o.destination WHERE o.matrix_kind = 'DEMAND' GROUP BY 1, 2), b AS (SELECT zo.sector_id AS o_sec, zd.sector_id AS d_sec, SUM(o.trips) AS trips FROM read_parquet('/home/user/CLAUDE/steam-ai/data/runs/BASE_2025_2025_52d15c27/tables/od.parquet') o JOIN read_parquet('/home/user/CLAUDE/steam-ai/data/runs/BASE_2025_2025_52d15c27/tables/zones.parquet') zo ON zo.zone_id = o.origin JOIN read_parquet('/home/user/CLAUDE/steam-ai/data/runs/BASE_2025_2025_52d15c27/tables/zones.parquet') zd ON zd.zone_id = o.destination WHERE o.matrix_kind = 'DEMAND' GROUP BY 1, 2) SELECT COALESCE(r.o_sec, b.o_sec) AS o_sec, COALESCE(r.d_sec, b.d_sec) AS d_sec, COALESCE(r.trips, 0) AS run_trips, COALESCE(b.trips, 0) AS base_trips, COALESCE(r.trips, 0) - COALESCE(b.trips, 0) AS delta, ABS(COALESCE(r.trips, 0) - COALESCE(b.trips, 0)) / NULLIF(b.trips, 0) AS shift_share FROM r FULL OUTER JOIN b USING (o_sec, d_sec) WHERE GREATEST(COALESCE(r.trips, 0), COALESCE(b.trips, 0)) >= 100.0 ORDER BY shift_share DESC NULLS FIRST, o_sec, d_sec

[src: findings/476781b9e9a2; od.csv (od) col trips; od.csv (od) col trips]

Likely cause: Land use, network or transit inputs changed in the sectors concerned, so the demand response is expected.

[src: findings/476781b9e9a2; od.csv (od) col trips; od.csv (od) col trips]

Suggested action: No action needed if the input changes were intended; the per-pair list is in the evidence.

[src: findings/476781b9e9a2; od.csv (od) col trips; od.csv (od) col trips]

Evidence

Metric	Value	Threshold
n_explained_shifts	24	
explained_pairs	24 items (full list in findings.json): origin_sector=S1, destination_sector=S4, run_trips=8,735, ... (7 keys), origin_sector=S4, destination_sector=S1, run_t...	
metric	trips	
sector_shift_share		10.0%
min_sector_pair_trips		100
explain_land_use_share		1.0%

[src: findings/476781b9e9a2; od.csv (od) col trips; od.csv (od) col trips]

Sources: od.csv (od) col trips; od.csv (od) col trips

[src: findings/476781b9e9a2; od.csv (od) col trips; od.csv (od) col trips]

Convergence and noise

Check status: ok; findings: 0; message: noise band written for 4870 link/sector-period rows.

[src: check_results.json]

The convergence check raised no findings.

[src: findings.json]

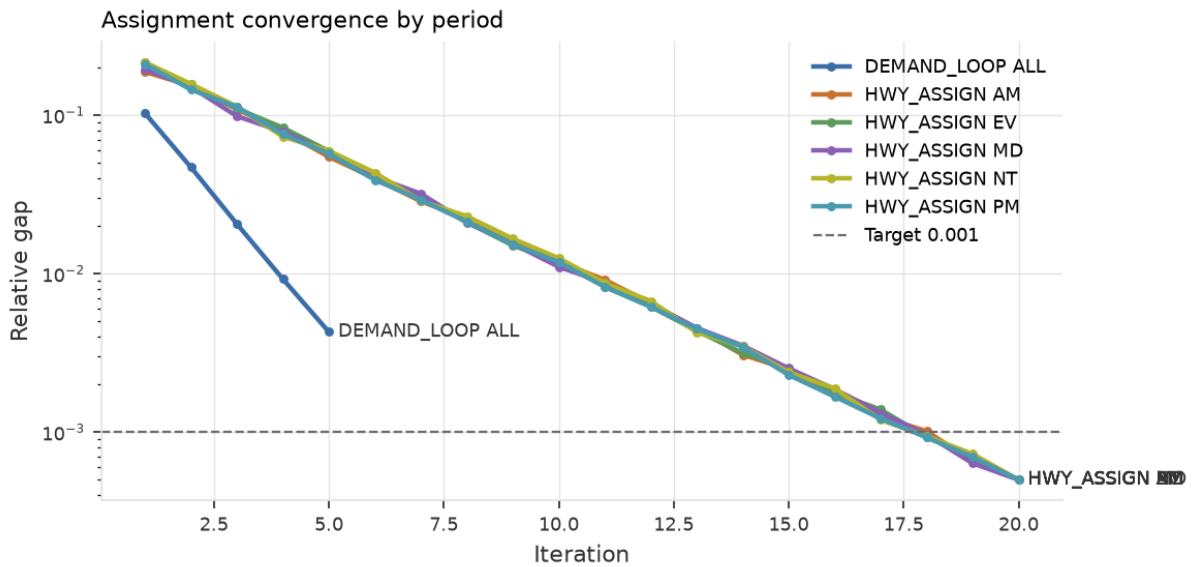
Final convergence values (relative gap target 0.001)

Stage	Period	Metric	Iterations	Final value	Target met
DEMAND_LOOP	ALL	REL_GAP	5	0.004283	no
HWY_ASSIGN	AM	GEH_PREV_ITER	20	0.2197	
HWY_ASSIGN	AM	MAX_FLOW_CHANGE	20	40.4	
HWY_ASSIGN	AM	REL_GAP	20	0.0005	yes
HWY_ASSIGN	EV	GEH_PREV_ITER	20	0.2197	
HWY_ASSIGN	EV	MAX_FLOW_CHANGE	20	20.8	
HWY_ASSIGN	EV	REL_GAP	20	0.0005	yes
HWY_ASSIGN	MD	GEH_PREV_ITER	20	0.2197	
HWY_ASSIGN	MD	MAX_FLOW_CHANGE	20	41.4	
HWY_ASSIGN	MD	REL_GAP	20	0.0005	yes
HWY_ASSIGN	NT	GEH_PREV_ITER	20	0.2197	
HWY_ASSIGN	NT	MAX_FLOW_CHANGE	20	13.0	
HWY_ASSIGN	NT	REL_GAP	20	0.0005	yes
HWY_ASSIGN	PM	GEH_PREV_ITER	20	0.2197	
HWY_ASSIGN	PM	MAX_FLOW_CHANGE	20	39.8	
HWY_ASSIGN	PM	REL_GAP	20	0.0005	yes

[src: convergence table; config/checks.yaml]

The noise band table has 4870 rows with columns level, location_id, period, metric, band_low, band_high, method.

[src: noise_band.parquet]



[src: convergence table (metric REL_GAP); config/checks.yaml rel_gap_target]

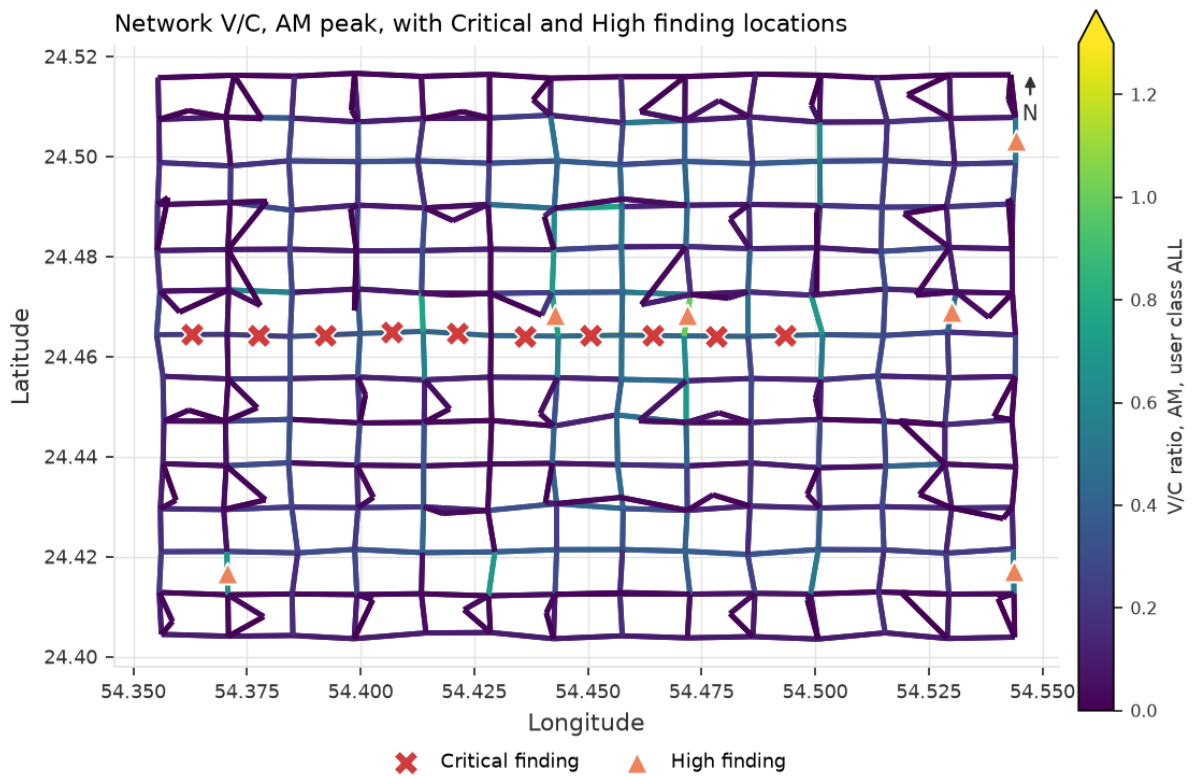
Key performance indicators

KPIs with their published definitions

KPI	Value	Period	Definition
Vehicle-km travelled, AM	2,091,840 veh-km	AM	Sum of link volume x length for user_class ALL in the AM period. SQL: SELECT SUM(f.volume * l.length_m / 1000.0) AS value FROM link_flows f JOIN links l USING (link_id) WHERE f.period = 'AM' AND f.user_class = 'ALL'
Share of links with V/C above 1.0, AM	1.3%	AM	Links with vc_ratio > 1.0 divided by all links carrying flow, AM, user_class ALL. SQL: SELECT AVG(CASE WHEN vc_ratio > 1.0 THEN 1.0 ELSE 0.0 END) AS value FROM link_flows WHERE period = 'AM' AND user_class = 'ALL' AND volume > 0
Volume-weighted congested speed, AM	74 km/h	AM	Sum(volume x congested speed) / Sum(volume) for AM, user_class ALL. SQL: SELECT SUM(volume * cong_speed_kph) / NULLIF(SUM(volume), 0) AS value FROM link_flows WHERE period = 'AM' AND user_class = 'ALL'
Total demand trips, AM	96,750 trips	AM	Sum of all DEMAND matrix cells in the AM period, all purposes and modes. SQL: SELECT SUM(trips) AS value FROM od WHERE matrix_kind = 'DEMAND' AND period = 'AM'
PT mode share, AM	14.3%	AM	PT demand trips divided by all demand trips, AM. SQL: SELECT SUM(CASE WHEN mode = 'PT' THEN trips ELSE 0 END) / NULLIF(SUM(trips), 0) AS value FROM od WHERE matrix_kind = 'DEMAND' AND period = 'AM'
Mean trip length, all periods	7.6 km	all	Trip-weighted mean of the DIST skim (metres, per data contract 8.3) over DEMAND matrix cells, converted to km. SQL: SELECT SUM(o.trips * s.value) / NULLIF(SUM(o.trips), 0) / 1000.0 AS value FROM od o JOIN skims s ON s.skim_kind = 'DIST' AND s.mode = o.mode AND s.period = o.period AND s.origin = o.origin AND s.destination = o.destination WHERE o.matrix_kind = 'DEMAND'
Road network length	1077.4 km	all	Sum of link length over all highway links (each direction counted). SQL: SELECT SUM(length_m) / 1000.0 AS value FROM links
Lane-km, all roads	1,809 lane-km	all	Sum of link length x lanes over all highway links. SQL: SELECT SUM(length_m * lanes) / 1000.0 AS value FROM links
Lane-km, freeways	606 lane-km	all	Sum of link length x lanes over links of class FWY. SQL: SELECT SUM(length_m * lanes) / 1000.0 AS value FROM links WHERE link_class = 'FWY'
Total demand trips, all periods	409,230 trips	all	Sum of all DEMAND matrix cells, every purpose, mode and period. SQL: SELECT SUM(trips) AS value FROM od WHERE matrix_kind = 'DEMAND'
Population	296,321 people	all	Sum of the POP land-use variable over all zones. SQL: SELECT SUM(value) AS value FROM land_use WHERE variable = 'POP'

[src: kpis.json; config/kpis.yaml]

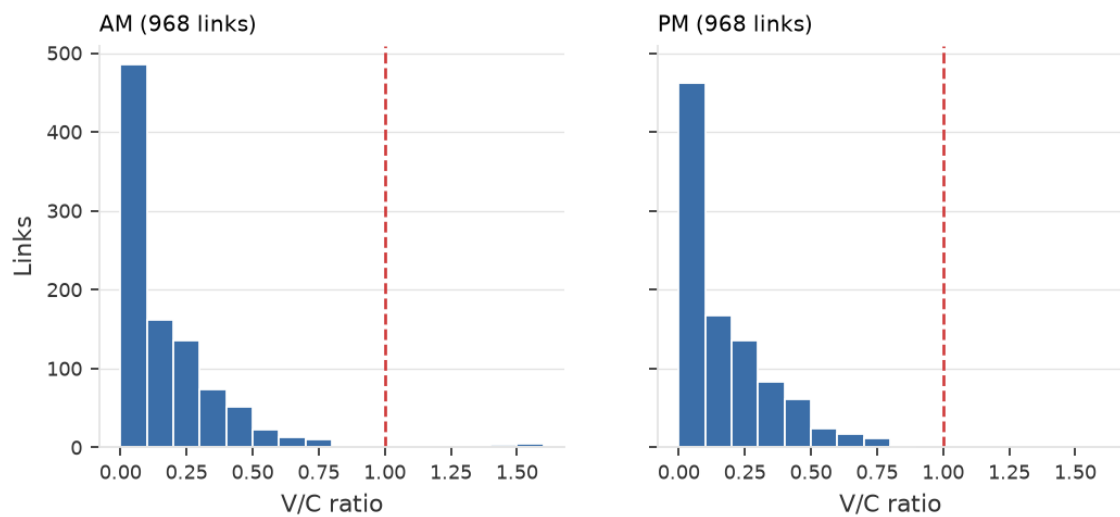
Maps and charts



Network map: links coloured by AM V/C (user class ALL), Critical and High finding locations marked

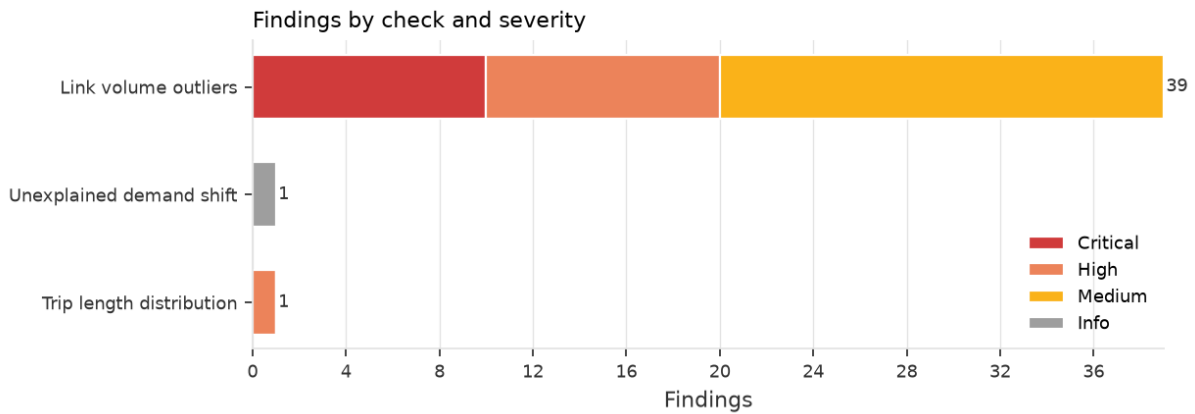
[src: links table (geometry_wkt); link_flows table period AM user_class ALL; findings.json]

Link V/C distribution, user class ALL



Distribution of link V/C ratios, AM and PM (user class ALL)

[src: link_flows table; user_class ALL; periods AM, PM]



Findings by check and severity

[src: findings.json; 41 findings]

Check coverage

13 checks were run: 13 completed, 0 skipped because a required table was missing, 0 ended in error.

[src: check_results.json]

Check coverage

Check	Status	Findings	Message	Duration (s)
Null values and identifier integrity	ok	0		0.35
Network connectivity	ok	0		0.02
Centroid connector placement	ok	0		0.01
Transit line integrity	ok	0		0.02
Land-use control totals	ok	0		0.01
Link volume outliers	ok	39		0.07
Convergence and noise band	ok	0	noise band written for 4870 link/sect or-period rows	0.04
Demand matrix sanity	ok	0		0.05
Trip length distribution	ok	1		0.22
Unrealistic speeds and times	ok	0		0.01
Parameter drift	ok	0		0.00
Unexplained demand shift	ok	1		0.04
Unused transit services	ok	0		0.02

[src: check_results.json]

Appendix

Method notes

Each check maps a computed score to a severity band. Bands are checked top-down and the first match wins. The thresholds below are read from config/checks.yaml (defaults: enabled=True, max_findings_per_check=200, group_issue_above=25, group_keep_worst=10).

[src: config/checks.yaml]

Check thresholds and severity rules

Check	Enabled	Parameters	Severity rules
null_and_id_integrity	yes	required_columns = {'links': ['link_id', 'a_node', 'b_node', 'length_m', 'link_class', 'capacity_vph', 'ffs_kph'], 'nodes': ['node_id', 'x', 'y'], 'zones': ['zone_id'], 'link_flows': ['link_id', 'period', 'user_class', 'volume']}	Critical when missing_node_refs > 0; Critical when missing_zone_refs > 0; High when missing_link_refs > 0; High when null_share > 0.01; Medium when null_share > 0
network_connectivity	yes	min_capacity_vph = 1; min_ffs_kph = 5; touch_tolerance_m = 1; near_gap_m = 50	Critical when issue == 'zero_capacity' or issue == 'zero_speed'; High when issue == 'node_id_mismatch'; Medium when issue == 'orphan_node'; High when issue == 'oneway_dead_end'; High when issue == 'disconnected_component'
centroid_connectors	yes	high_class_links = FWY, EXP; connector_classes = CONN; max_connector_length_m = 5,000	High when issue == 'connector_on_high_class'; Medium when issue == 'connector_too_long'
transit_line_integrity	yes	headway_min_range = 2, 120; modes_headway_max = {'METRO': 20, 'LRT': 30, 'BRT': 30, 'BUS': 120, 'RAIL': 120, 'TRAM': 30, 'FERRY': 120}	Critical when issue == 'broken_sequence'; High when issue == 'headway_out_of_range'; Critical when issue == 'node_not_in_network'
land_use_control_totals	yes	tolerance_share = 0.02	Critical when abs_diff_share > 0.10; High when abs_diff_share > 0.05; Medium when abs_diff_share > 0.02
link_volume_outliers	yes	vc_critical = 1.3; vc_high = 1.1; vc_medium = 0.95; zscore_threshold = 3; min_vc_for_outlier = 0.25; min_group_size = 20; low_volume_share_of_group_median = 0.02; user_class = ALL; periods = AM, PM; exclude_link_classes = CONN	Critical when vc_ratio >= 1.30; High when vc_ratio >= 1.10; High when abs_zscore >= 4.0; Medium when vc_ratio >= 0.95 or abs_zscore >= 3.0; Medium when too_low
convergence_and_noise	yes	rel_gap_target = 0.001; rel_gap_high = 0.01; geh_stable = 1; geh_stable_share_target = 0.95; noise_band_iterations = 3; stages = HWY_ASSIGN, PT_ASSIGN	Critical when rel_gap > 0.01; High when rel_gap > 0.001; Medium when stable_share < 0.95
matrix_sanity	yes	intrazonal_share_max = 0.15; row_col_ratio_max = 5; min_zone_total_for_ratio = 50; pa_format_share = 0.25; fractional_cell_max_trips = 0.01; fractional_share_min = 0.01	Critical when issue == 'negative_cells'; High when issue == 'row_col_imbalance'; Medium when issue == 'intrazonal_share'; Info when issue == 'fractional_artefact'
trip_length_distribution	yes	max_mean_diff_share = 0.15; max_mean_diff_share_high = 0.3; distance_bins_km = 0, 2, 5, 10, 20, 40, 80, 200; skim_dist_units = km; reference_mean_km = {}	High when mean_diff_share > 0.30; Medium when mean_diff_share > 0.15
unrealistic_speeds_times	yes	min_cong_speed_kph = 5; max_speed_over_ffs_ratio = 1.05; max_delay_s = 900; periods = AM, PM; user_class = ALL	High when issue == 'speed_above_ffs'; High when issue == 'speed_below_floor'; Medium when issue == 'delay_outlier'
parameter_drift	yes	numeric_tolerance = 1e-09	High when drifted
unexplained_demand_and_shift	yes	sector_shift_share = 0.1; catchment_km = 5; min_sector_pair_trips = 100; explain_land_use_share = 0.01	High when shift_share > 0.25; Medium when shift_share > 0.10
unused_transit_services	yes	min_load_factor = 0.05; min_catchment_demand = 500; catchment_km = 1; pt_modes = PT	High when load_factor < 0.02; Medium when load_factor < 0.05

[src: config/checks.yaml]

Glossary

Glossary

Term	Meaning
V/C	Volume-to-capacity ratio: link volume per hour divided by link capacity per hour. Values above 1.0 mean demand exceeds capacity.
GEH	A statistic comparing two flows, M and C: $\sqrt{2 (M - C)^2 / (M + C)}$. Values below 5 are usually treated as a good match.
Relative gap	Assignment convergence measure: the excess cost of the current route choices over the cheapest possible routes, divided by total cost. Smaller is better.
Noise band	The size of link flow change between the last assignment iterations. Differences inside the band are not treated as significant.
Centroid connector	An artificial link that loads a zone's trips onto the network at a chosen node.
Screenline	An imaginary line across the network on which modelled and observed crossings are compared.

[src: STEAM-AI glossary]

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